

Bitcoin Sentiment and Trader Behavior Analysis

Methodology

The Bitcoin Fear & Greed Index dataset was merged with Hyperliquid trading records using a daily date alignment strategy. SQL Server was used for data preparation, cleaning, timestamp conversion, duplicate analysis, and metric generation. Python was then used for exploratory analysis, visualization, segmentation, predictive modeling, and clustering.

Key metrics generated:

- Daily PnL
- Win Rate
- Average Trade Size
- Trades Per Day
- Trading Bias
- Long/Short Counts

Key Findings

1. Fear Days Produced Highest Average Profitability

Average Daily PnL:

- Fear: \$209,372
- Greed: \$99,676
- Extreme Greed: \$35,393
- Neutral: \$19,842

This suggests traders captured larger opportunities during fearful market conditions.

2. Trading Activity Increased During Fear Periods

Average Trades Per Day:

- Fear: 4,183
- Greed: 1,134
- Extreme Greed: 1,392
- Neutral: 893

Market uncertainty appears to encourage more active trading.

3. Larger Position Sizes During Fear and Greed

Average Position Size:

- Fear: \$5,926
- Greed: \$5,889
- Extreme Greed: \$4,344
- Neutral: \$3,793

Traders were willing to deploy more capital during emotionally driven market phases.

Strategy Recommendations

Strategy 1: Aggressive Opportunity Capture During Fear

When sentiment shifts to Fear:

- Increase market monitoring
- Allow higher trading frequency
- Focus on short-term volatility opportunities

Strategy 2: Risk Control During Greed

During Greed periods:

- Reduce position concentration
- Limit leverage exposure
- Prioritize risk management over aggressive expansion

Advanced Analytics

A Random Forest classification model was built to predict trader profitability using sentiment and behavioral metrics.

K-Means clustering identified trader archetypes:

1. Active High-Risk Traders
2. Balanced Traders
3. Conservative Low-Frequency Traders

An interactive Streamlit dashboard was also developed for exploratory analysis.